- Observe $X_1, \dots, X_n \sim P$ for $P \in \mathcal{P}$. ($\mathcal{P}$: statistical model.)
- $\psi(P)$ for $\psi: \mathcal{P} \to \mathbb{R}^k$ is the quantity of interest.

Q: How to estimate $\psi(P)$ "efficiently"

- For this we need the tangent set: take path $t \mapsto P_t$ with $P_0 = P$ and score

$$g = \frac{d}{dt} \log dP_t \Big|_{t=0}.$$



Set of scores is a tangent set $\dot{\mathcal{P}}_P$ at P.

- Assume ψ is differentiable in the sense that

$$\frac{\psi(P_t) - \psi(P)}{t} \to P \tilde{\psi}_P g$$

$\forall t \mapsto P_t$ with score g.

- $\tilde{\psi}_P \in \overline{\text{lin}}\, \dot{\mathcal{P}}_P$ is the efficient influence function.
- $P \tilde{\psi}_P \tilde{\psi}_P^T$ "lower bound on variance."

Talk about finding any influence function and projecting it onto the tangent space!

example: parametric model.

$$\mathcal{P} = \{P_\theta : \theta \in \Theta\}, \quad \Theta \subseteq \mathbb{R}^m \text{ open}$$

s.t. P_θ has density p_θ w.r.t. a σ-finite μ.

$$\theta \mapsto \dot{\ell}_\theta := \frac{d}{d\theta} \log p_\theta.$$

Submodel $t \mapsto P_{\theta + th}$, $h \in \mathbb{R}^m$ gives a score $h^T \dot{\ell}_\theta$:

$$\frac{d}{dt} \log dP_{\theta+th} = \frac{d}{dt} \ell_{\theta+th} = h^T \dot{\ell}_{\theta+th}$$

So tangent set $\{h^T \dot{\ell}_\theta \mid h \in \mathbb{R}^m\}$.

Let $\chi: \Theta \to \mathbb{R}^k$ be differentiable. Set $\psi(P_\theta) := \chi(\theta)$, then

$$\lim_{t \downarrow 0} \frac{\psi(P_{\theta+th}) - \psi(P_\theta)}{t} = \dot{\chi}_\theta h$$

($\dot{\chi}_\theta$: Jacobian at θ.)

If Fisher information matrix $I_\theta := P_\theta \dot{\ell}_\theta \dot{\ell}_\theta^T$ is invertible,

$$\dot{\chi}_\theta h = \dot{\chi}_\theta I_\theta^{-1} I_\theta h = \dot{\chi}_\theta I_\theta^{-1} P_\theta \dot{\ell}_\theta \dot{\ell}_\theta^T h$$

$$= P_\theta \underbrace{\dot{\chi}_\theta I_\theta^{-1} \dot{\ell}_\theta}_{=: \tilde{\psi}_{P_\theta}} \dot{\ell}_\theta^T h$$

Q: Is it the efficient IF?

$$P_\theta \tilde{\psi}_{P_\theta} \tilde{\psi}_{P_\theta}^T = P_\theta \dot{\chi}_\theta I_\theta^{-1} \dot{\ell}_\theta \dot{\ell}_\theta^T I_\theta^{-1} \dot{\chi}_\theta^T$$

$$= \dot{\chi}_\theta I_\theta^{-1} \dot{\chi}_\theta^T$$

Double-robust estimator.

- Treatment A in $\{0,1\}$
- Potential outcome Y^a, $a=0,1$.
- covariates L.

- Model consists of all square integrable laws for $(Y^0 \mathbb{1}\{A=0\}, Y^1 \mathbb{1}\{A=1\}, A, L)$ s.t.
  - (C): $Y^A = Y$
  - (CE): $Y^a \perp\!\!\!\perp A \mid L$ $a=0,1$.
  - (P^a): $0 < \xi(a|L) := \Pr(A=a|L)$

- Assume $Y^0, Y^1 \mid L=\ell$ has density $(y^0,y^1) \mapsto p(y^0,y^1|\ell)$, with marginals $y^0 \mapsto p^0(y^0|\ell)$, $y^1 \mapsto p^1(y^1|\ell)$.

- Assume L has density p^L.

So full model is parametrised by (ξ, p, p^L) and write $P_{(\xi,p,p^L)}$ as the law.
The likelihood factorizes as

$$\begin{aligned}(\xi,p,p^L) \mapsto & \xi(1|L)^A (1-\xi(1|L))^{1-A} \\ & \cdot p^L(L) \cdot p^0(Y^0 \mathbb{1}_{A=0} | L) \\ & \cdot p^1(Y^1 \mathbb{1}_{A=1} | L) \\ = & \xi(1|L)^A (1-\xi(1|L))^{1-A} \\ & p^L(L)\, p^0(Y|L)^{1-A} \\ & p^1(Y|L)^A\end{aligned}$$

$$p(y^0, y^1, \ell, a) = p^{a|Y^1, Y^0, L} \cdot p^{Y^1, Y^0|L}\, p^L = \xi(a|\ell)\, p(y^0,y^1|\ell)\, p^L(\ell)$$

Use submodels of the form
- $\xi_t(1|\ell) = \xi(1|\ell) + t\phi(\ell)$

- $p^0_t(y|\ell) = (1+t\beta^0(y|\ell))\, p^0(y|\ell)$ s.t.
$$\int \beta^0(y|\ell)\, p^0(y|\ell)\, dy = 0$$

- $p^1_t(y|\ell) = (1+t\beta^1(y|\ell))\, p^1(y|\ell)$ s.t.
$$\int \beta^1(y|\ell)\, p^1(y|\ell)\, dy = 0$$

- $p^L_t = (1+t\gamma)\, p^L$ s.t. $\int \gamma(\ell)\, p^L(\ell)\, d\ell = 0$.

Also assume $\beta^0, \beta^1, \gamma, \phi$ are bounded.

Fixing all submodel parts except one gives

ξ-score: $\dfrac{\mathbb{1}_{A=1} - \xi(1|L)}{\xi(0|L)\xi(1|L)}\phi(L)$

p^0-score: $\mathbb{1}_{A=0}\beta^0(Y|L)$

p^1-score: $\mathbb{1}_{A=1}\beta^1(Y|L)$

p^L-score: $\gamma(L)$

The ATE is a functional of the parameter:

$$\begin{aligned}\chi(\xi,p,p^L) &= \iint y\,(p^1(y|\ell) - p^0(y|\ell))\, p^L(\ell)\, dy\, d\ell \\ &= \int \mu^1(\ell) - \mu^0(\ell)\, p^L(\ell)\, d\ell,\end{aligned}$$

with $\mu^a(\ell) = E(Y^a | L=\ell)$

$$\begin{aligned}E\,Y^1 - Y^0 &= E_L E(Y^1 - Y^0 | L) \\ &= E_L\left(\int y\, p^1(y|L)\, dy - \int y\, p^0(y|L)\, dy\right)\end{aligned}$$

Write $X = (\mathbb{1}_{A=0} Y^0, \mathbb{1}_{A=1} Y^1, A, L)$.

Set $\chi^a(\xi,p,p^L) = \int\int y\, p^a(y|\ell)\, p^L(\ell)\, dy\, d\ell$

Then

$$\begin{aligned}\frac{d}{dt}\Big|_{t=0} \chi^a(\xi_t,p_t,p^L_t) &= \iint \frac{d}{dt}\Big|_{t=0} y\,(1+t\beta^a(y|\ell)) \\ &\quad \cdot p^a(y|\ell)(1+t\gamma(\ell))\, p^L(\ell)\, dy\, d\ell \\ &= \iint y\,\beta^a(y|\ell)\, p^a(y|\ell)\, p^L(\ell) + y\, p^a(y|\ell)\,\gamma(\ell)\, p^L(\ell)\ dy\,d\ell \\ &= \iint y\,(\beta^a(y|\ell) + \gamma(\ell))\, p^a(y|\ell)\, p^L(\ell)\, dy\, d\ell\end{aligned}$$

$$= E_L E\left(y^a(\beta^a(y^a|L) + \gamma(L))\,|\,L\right)$$

$$= E_L E(y^a \beta^a(y^a|L)|L) + E_L \gamma(L) E(y^a|L) \quad = (*)$$

A score q is of the form

$$q = q_{\beta,\gamma} + \mathbb{1}_{A=0}\beta^0(y|L) + \mathbb{1}_{A=1}\beta^1(y|L) + \gamma(L)$$

Note, $E_L E(y^a \beta^a(y^a|L)|L)$

$$= E_L E\left(y^a \beta^a(y|L)|L, A=a\right)$$

$$= E_L \frac{E(\mathbb{1}_{A=a}\beta^a(y|L)y^a|L)}{\xi(a|L)}$$

$$= E\left(\underbrace{\mathbb{1}_{A=a}\beta^a(y|L)}_{\beta^a\text{-score}} \cdot \underbrace{\frac{\mathbb{1}_{A=a}y^a}{\xi(a|L)}}_{\text{not mean zero}}\right)$$

not mean zero is inverse propensity Weighting formula.

Centering

Note

$$E(\mathbb{1}_{A=a}\beta^a(y|L)\cdot\mu^a(L)/\xi(a|L))$$

$$= E_L E(\beta^a(y^a|L)\mu^a(L)/\xi(a|L)|L)$$

$$= E_L \frac{\mu^a(L)}{\xi(a|L)} E(\beta^a(y^a|L)|L) = 0$$

and hence $(*)$

$$= E\left(\frac{y^a - \mu^a(L)}{\xi(a|L)} \cdot \mathbb{1}_{A=a}\beta^a(y|L)\right) + E_L\left(\gamma(L)(\mu^a(L) - \chi^a)\right)$$

Set $\tilde{\chi}^a_{\xi,p,p^L}(X) = \frac{\mathbb{1}_{A=a}}{\xi(a|L)}(y^a - \mu^a(L)) + \mu^a(L) - \chi^a(\xi,p,p^L)$

Q: does above equal

$$E\,\tilde{\chi}^a_{\xi,p,p^L}(X)\,q\,?$$

$\frac{\mathbb{1}_{A=a}}{\xi(a|L)}(y^a - \mu^a(L))$ has inner product zero with $p^{a'}$-score, $a' \neq a$. Also,

$$E\left(\frac{\mathbb{1}_{A=a}}{\xi(a|L)}(y^a - \mu^a(L))\,\middle|\,A,L\right) = 0,$$

$\Rightarrow$ inner product zero with ξ-score and p^L-score.

Also,

$$E(\mu^a(L) - \chi^a)q$$

$$= E_L(\mu^a(L) - \chi^a)E(q|L)$$

$$= E\,\gamma(L)(\mu^a(L) - \chi^a).$$

Final check: Is the IF in the tangent space? Yes.

The EIF is given by

$$\tilde{\chi}_{\xi,p,p^L}(X) = \frac{\mathbb{1}_{A=1}}{\xi(1|L)}(y^1 - \mu^1(L)) - \frac{\mathbb{1}_{A=0}}{\xi(1|L)}(y^0 - \mu^0(L)) + (\mu^1 - \mu^0)(L) - \chi(\xi,p,p^L)$$

An estimator T_n is efficient iff

$$T_n = \chi(\xi,p,p^L) + \mathbb{P}_n\tilde{\chi}_{\xi,p,p^L}(X_i) + o_p(1/\sqrt{n})$$

So for estimators $\hat{\xi}_n, \hat{p}_n, \hat{p}^L_n$, set

$$T_n = \chi(\hat{\xi}_n,\hat{p}_n,\hat{p}^L_n) + \mathbb{P}_n\tilde{\chi}_{\hat{\xi}_n,\hat{p}_n,\hat{p}^L_n}(X_i)$$

$$= \mathbb{P}_n\left(\frac{\mathbb{1}_{A=1}}{\hat{\xi}_n(1|L)}(y^1 - \hat{\mu}^1_n(L)) - \frac{\mathbb{1}_{A=0}}{\hat{\xi}_n(0|L)}(y^0 - \hat{\mu}^0_n(L)) + (\hat{\mu}^1_n - \hat{\mu}^0_n)(L)\right),$$

which is the double robust estimator.

Note EIF is contained in the p-, p^L-score space. So if ξ would be known, or partially known, the efficiency bound does not change.